

Factorization Exercises — Solutions

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Part 1 Answers

1. true
2. false
3. true
4. false
5. true
6. true
7. true
8. false
9. false
10. true

Part 2 Answers

1. $A = 3, B = 8$
2. $A = 2, B = -13$
3. $A = 2, B = 7$
4. $A = 5, B = 1$
5. $A = 5, B = 3$
6. $A = -7, B = 21$
7. $A = 3, B = 12$
8. $A = 2, B = 1$

Part 3 Answers

1. $x^2 - 3x - 28$
2. $2x^2 + 7x - 15$
3. $12x^2 - 11x + 2$
4. $-8x^2 + 14x + 15$
5. $x^2 + 12x + 36$
6. $4x^2 - 20x + 25$
7. $16x^2 + 24x + 9$
8. $9x^2 - 1$
9. $36x^2 - 4$
10. $2x^3 + 4x^2 - 6x - 12$
11. $2x^3 - 13x^2 + 19x - 20$
12. $7x^2 - 19x + 8$

Part 4 Answers

1. 9604
2. 10609
3. 1400
4. 39984
5. 2450.25
6. 10000
7. 1000000
8. 2025

1 Level2 Answers

Part1 Answers

1. true
2. false
3. true
4. false
5. false
6. true
7. true
8. false
9. false
10. true

Part2 Answers

1. $A = \frac{5}{2}, B = -19$
2. $A = 3, B = 26$
3. $A = 2, B = -8$
4. $A = 7, B = -6$
5. $A = -8, B = 21$
6. $A = 7, B = -1$
7. $A = 4, B = 9$
8. $A = 4, B = 6$

Part3 Answers

1. $6x^2 - 7x - 20$
2. $8x^2 - 22x - 6$
3. $10x^2 - 41x + 21$
4. $-6x^2 - 7x + 20$
5. $9x^2 - 24x + 16$
6. $25x^2 + 20x + 4$
7. $4x^2 - 28x + 49$
8. $16x^2 - 25$
9. $9x^2 - 64$
10. $3x^3 + 5x^2 - 17x + 5$
11. $6x^3 - 13x^2 + 19x - 15$
12. $6x^2 + 27x + 7$

Part4 Answers

1. 38809
2. 42436
3. 3600
4. 89951
5. 3660.25
6. 40000
7. 3996009
8. 3040

Part5 Answers

1. $(6x^2 + 7x - 20) \text{ cm}^2$
2. $(16a^2 - 40a + 25) \text{ cm}^2$
3. 157
4. $x(x + 7) = 30 \implies x^2 + 7x - 30 = 0$
5. $(3x + 4)$
6. $(2k + 1)^2 - (2k - 1)^2 = 8k$, $8k$ is divisible by 8.